

FIT3182 Evidence

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Table of Contents

Logged Drop Pairs	2
Sleep Time Window Justification	3
Output for Producers	4
Output for Streaming/Joining	5
Chart Images	6
Chart 1	6
Chart 2	6
Chart 3	7
Chart 4	7

Logged Drop Pairs

Below is evidence of drop pairs, we have set the join window to 30 seconds for testing purposes, car_plate = UX 3296 has been flagged for having a total time of more than 30 seconds. Therefore, joining would be skipped and dropped, and the db does not save the average violation for this pair.

If we look at the timestamp for camera_id 2 - 44:05 and camera_id 3 - 44:38, which gives us 33 seconds between them (109.09 km/h). Although this is an average violation, since we have set the join window to 30 seconds, it won't appear in the database because of this.

```
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED - reason: unmatched A→B (cam 1 → cam 2) - no partner event within join window (30 seconds)
car_plate=UX 3296 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 10:43:32 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
```

```
fit3182_db> db.violation.find( { car_plate: "UX 3296" })
[
  {
    _id: ObjectId('6a142d248963cd6a0dab1449'),
    date: ISODate('2024-01-01T00:00:00.000Z'),
    car_plate: 'UX 3296',
    violations: [
      {
        violation_type: 'INSTANTANEOUS',
        camera_id_start: 1,
        camera_id_end: 1,
        timestamp_start: ISODate('2024-01-01T10:43:32.000Z'),
        timestamp_end: ISODate('2024-01-01T10:43:32.000Z'),
        speed_reading: 110.8,
        avg_speed: 110.8
      },
      {
        violation_type: 'INSTANTANEOUS',
        camera_id_start: 2,
        camera_id_end: 2,
        timestamp_start: ISODate('2024-01-01T10:44:05.747Z'),
        timestamp_end: ISODate('2024-01-01T10:44:05.747Z'),
        speed_reading: 110.5,
        avg_speed: 110.5
      },
      {
        violation_type: 'INSTANTANEOUS',
        camera_id_start: 3,
        camera_id_end: 3,
        timestamp_start: ISODate('2024-01-01T10:44:38.021Z'),
        timestamp_end: ISODate('2024-01-01T10:44:38.021Z'),
        speed_reading: 108.1,
        avg_speed: 108.1
      }
    ]
  }
]
fit3182_db> 
```

Sleep Time Window Justification

Producer A = 5 seconds

Producer B = 0.470779 seconds

Producer C = 0.335227

For formula calculation of each sleep time for the producers are:

Producer A = 5 seconds

Producer B = (Total duration between last data and first data in the camera_event_A.csv / number of batches) / (Total duration between last data and first data in the camera_event_A.csv / number of batches) * 5 seconds

Producer B = (Total duration between last data and first data in the camera_event_B.csv / number of batches) / (Total duration between last data and first data in the camera_event_A.csv / number of batches) * 5 seconds

Justification:

In order to match timings of each batch, reduce the amount of unnecessary drop pairs from watermark.

Output for Producers

Publishing 556340 records across 27817 batch(es) to topic 'camera-events-A'
Sleep between batches: 5s

— Batch 1 (20 records) —

```
Published: {'event_id': 'd40c586c-5be6-4743-a1e3-2269d9edaa72', 'batch_id': 1, 'car_plate': 'KRN 7', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:04', 'speed_reading': 77.2, 'producer_id': 'A'}
Published: {'event_id': '85c08e3c-a0b5-45d8-a79c-df8f9a6d5829', 'batch_id': 1, 'car_plate': 'ICE 8', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:05', 'speed_reading': 103.7, 'producer_id': 'A'}
Published: {'event_id': 'f5834b79-771b-4931-8da2-a5ad7f4ccd02', 'batch_id': 1, 'car_plate': 'QE 1820', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:03', 'speed_reading': 67.4, 'producer_id': 'A'}
Published: {'event_id': 'd0e547bb-c4a7-4750-b7b4-8076e9b47f4f', 'batch_id': 1, 'car_plate': 'CJM 924', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:01', 'speed_reading': 148.3, 'producer_id': 'A'}
Published: {'event_id': 'f3162606-1b2e-407f-951d-61d14c0a7b09', 'batch_id': 1, 'car_plate': 'CJP 278', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:02', 'speed_reading': 125.2, 'producer_id': 'A'}
Published: {'event_id': 'c69852f7-cd9a-4892-b225-8f8a36ec017b', 'batch_id': 1, 'car_plate': 'ZPG 90', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:01', 'speed_reading': 95.7, 'producer_id': 'A'}
Published: {'event_id': '5113d990-4da6-41d2-b292-2516e7d8dc07', 'batch_id': 1, 'car_plate': 'KKE 15', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:00', 'speed_reading': 150.3, 'producer_id': 'A'}
Published: {'event_id': '67ce0672-95b5-43bd-9940-2e16db5f747a', 'batch_id': 1, 'car_plate': 'WJX 32', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:01', 'speed_reading': 97.4, 'producer_id': 'A'}
Published: {'event_id': '7c5c112b-7521-4bbc-99f8-3c3a85cc3fe1', 'batch_id': 1, 'car_plate': 'FO 32', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:05', 'speed_reading': 99.7, 'producer_id': 'A'}
Published: {'event_id': '213f0627-2be4-409c-972b-669627ca0461', 'batch_id': 1, 'car_plate': 'FA 35', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:03', 'speed_reading': 81.7, 'producer_id': 'A'}
Published: {'event_id': '5a449c97-5e3b-4470-a20c-8eef61019732', 'batch_id': 1, 'car_plate': 'CZ 592', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:02', 'speed_reading': 154.9, 'producer_id': 'A'}
Published: {'event_id': 'b57a0e44-b6e5-421e-bb5b-a4e1bb704bae0', 'batch_id': 1, 'car_plate': 'UIT 229', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:00', 'speed_reading': 140.9, 'producer_id': 'A'}
Published: {'event_id': 'c6965fdc-074f-4008-a8d3-f60f6bc4d223', 'batch_id': 1, 'car_plate': 'AH 8', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:04', 'speed_reading': 159.5, 'producer_id': 'A'}
Published: {'event_id': '3306546c-d263-43f2-9d8c-87732e10e099', 'batch_id': 1, 'car_plate': 'CX 6145', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:05', 'speed_reading': 105.0, 'producer_id': 'A'}
Published: {'event_id': 'f108727e-e97d-4d8e-b109-e7e763b64398', 'batch_id': 1, 'car_plate': 'FZ 766', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:05', 'speed_reading': 132.9, 'producer_id': 'A'}
Published: {'event_id': '7214629f-2fbb-4787-b953-e163720a6c61', 'batch_id': 1, 'car_plate': 'KZE 5', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:01', 'speed_reading': 133.3, 'producer_id': 'A'}
Published: {'event_id': '3ca5e275-0cf6-43a0-8c28-65040c608b02', 'batch_id': 1, 'car_plate': 'HP 1', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:03', 'speed_reading': 140.0, 'producer_id': 'A'}
Published: {'event_id': '562feec6-d50b-4207-b2ea-e3ba868f20f9', 'batch_id': 1, 'car_plate': 'WC 89', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:03', 'speed_reading': 96.7, 'producer_id': 'A'}
Published: {'event_id': '25a3e2ad-502e-4273-9062-6f00ec63f62c', 'batch_id': 1, 'car_plate': 'EOT 00', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:04', 'speed_reading': 142.6, 'producer_id': 'A'}
Published: {'event_id': '0abc5099-64d1-4b1a-b623-084b19ce9e4e', 'batch_id': 1, 'car_plate': 'WA 0712', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:04', 'speed_reading': 142.1, 'producer_id': 'A'}
Batch 1 complete. Sleeping 5s...
```

— Batch 2 (20 records) —

```
Published: {'event_id': 'be770f2f-e15a-463c-b4a7-955ee3e14924', 'batch_id': 2, 'car_plate': 'WB 418', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:01', 'speed_reading': 148.7, 'producer_id': 'A'}
Published: {'event_id': '3930bb19-483f-4f01-a6b2-c5e757d1cc53', 'batch_id': 2, 'car_plate': 'QJ 53', 'camera_id': 1, 'timestamp': '2024-01-01T08:00:02', 'speed_reading': 82.6, 'producer_id': 'A'}
...
Published: {'event_id': 'e481f1d9-0eae-400c-adce-fa6851933bf1', 'batch_id': 24, 'car_plate': 'DD5 5', 'camera_id': 1, 'timestamp': '2024-01-01T10:58:33', 'speed_reading': 126.4, 'producer_id': 'A'}
Published: {'event_id': '920e147c-4a04-405c-95cf-1219bc157ce4', 'batch_id': 24, 'car_plate': 'VQ 75', 'camera_id': 1, 'timestamp': '2024-01-01T10:58:36', 'speed_reading': 74.1, 'producer_id': 'A'}
Batch 24 complete. Sleeping 5s...
```

2m 19.1s

Python

Connected to Kafka at 192.168.56.1:9092

Publishing 556340 records across 294206 batches
Sleep between batches: 0.470779s

Batch 1 (5 records)

Published: {'event_id': 'b5b7dae7-3bf9-4f75-aea8-398d3bde3a41', 'batch_id': 1, 'car_plate': 'CJM 924', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:26.029536', 'speed_reading': 156.8, 'producer_id': 'A'}
Published: {'event_id': '1649c03b-cd6f-47bd-b4f2-c44fcf60524d', 'batch_id': 1, 'car_plate': 'CZ 592', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:26.729115', 'speed_reading': 163.9, 'producer_id': 'A'}
Published: {'event_id': 'a9c6a4b7-a6b6-4bd7-9719-88bd895814f4', 'batch_id': 1, 'car_plate': 'UIT 229', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:26.887170', 'speed_reading': 134.1, 'producer_id': 'A'}
Published: {'event_id': '4e13ec51-02db-442c-b7aa-3a2050dd1adi', 'batch_id': 1, 'car_plate': 'WA 0712', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:27.601617', 'speed_reading': 133.8, 'producer_id': 'A'}
Published: {'event_id': '74248dd9-c9f8-4ad2-94fb-dc5610a0f735', 'batch_id': 1, 'car_plate': 'AH 8', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:28.125701', 'speed_reading': 158.9, 'producer_id': 'A'}
Batch 1 complete. Sleeping 0.470779s...

Batch 2 (1 records)

Published: {'event_id': 'c14b050b-9319-4705-9adc-74c99eff0791', 'batch_id': 2, 'car_plate': 'FZ 766', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:33.114696', 'speed_reading': 136.0, 'producer_id': 'A'}
Batch 2 complete. Sleeping 0.470779s...

Batch 3 (1 records)

Published: {'event_id': 'e5ae9cab-569c-45e0-b31f-9e6f07d55838', 'batch_id': 3, 'car_plate': 'WC 89', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:41.774758', 'speed_reading': 93.5, 'producer_id': 'A'}
Batch 3 complete. Sleeping 0.470779s...

Batch 4 (4 records)

Published: {'event_id': '9f2e4a15-24fb-4879-9129-5800338088fa', 'batch_id': 4, 'car_plate': 'EOT 00', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:28.290929', 'speed_reading': 138.8, 'producer_id': 'A'}
Published: {'event_id': 'ee787113-e0bd-49d6-9cf9-054509625449', 'batch_id': 4, 'car_plate': 'HP 1', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:28.740321', 'speed_reading': 142.8, 'producer_id': 'A'}
Published: {'event_id': 'cada35fc-33a8-4b0f-a640-e0ddfc2a73df', 'batch_id': 4, 'car_plate': 'KZE 5', 'camera_id': 2, 'timestamp': '2024-01-01T08:00:29.186382', 'speed_reading': 135.2, 'producer_id': 'A'}
...
Published: {'event_id': '1be34164-25d7-42ef-9ed4-b04390b2a0a4', 'batch_id': 281, 'car_plate': 'WVR 8620', 'camera_id': 2, 'timestamp': '2024-01-01T11:28:00.660166', 'speed_reading': 136.6, 'producer_id': 'A'}
Published: {'event_id': 'd73132de-ebca-43d4-9bb9-ccc4c4f45b5b', 'batch_id': 281, 'car_plate': 'UXB 4819', 'camera_id': 2, 'timestamp': '2024-01-01T11:28:09.894777', 'speed_reading': 125.8, 'producer_id': 'A'}
Batch 281 complete. Sleeping 0.470779s...

Connected to Kafka at 192.168.56.1:9092

Publishing 556340 records across 414871 batches
Sleep between batches: 0.335227s

Batch 1 (1 records)

Published: {'event_id': 'd86e8cdb-c387-4ccb-b35a-346302238824', 'batch_id': 1, 'car_plate': 'UIT 229', 'camera_id': 3, 'timestamp': '2024-01-01T08:00:54.958092', 'speed_reading': 130.8, 'producer_id': 'A'}
Batch 1 complete. Sleeping 0.335227s...

Batch 2 (1 records)

Published: {'event_id': 'f804c4e8-022d-4243-b43e-9ec0806ce0ff', 'batch_id': 2, 'car_plate': 'CJM 924', 'camera_id': 3, 'timestamp': '2024-01-01T08:00:47.168827', 'speed_reading': 164.5, 'producer_id': 'A'}
Batch 2 complete. Sleeping 0.335227s...

Batch 3 (2 records)

Published: {'event_id': 'a2a0b391-fb7c-44a3-8f8b-ddf40bba3d05', 'batch_id': 3, 'car_plate': 'WA 0712', 'camera_id': 3, 'timestamp': '2024-01-01T08:00:53.815270', 'speed_reading': 134.4, 'producer_id': 'A'}
Published: {'event_id': 'e0daa388-6036-4b7d-ba58-bed0808c1611', 'batch_id': 3, 'car_plate': 'KZE 5', 'camera_id': 3, 'timestamp': '2024-01-01T08:00:57.613888', 'speed_reading': 125.3, 'producer_id': 'A'}
Batch 3 complete. Sleeping 0.335227s...

Batch 4 (2 records)

Published: {'event_id': '55a60e1c-818e-4e55-bc02-0627765629b2', 'batch_id': 4, 'car_plate': 'ICE 8', 'camera_id': 3, 'timestamp': '2024-01-01T08:01:16.405088', 'speed_reading': 108.0, 'producer_id': 'A'}
Published: {'event_id': 'daaf0000-91fa-4ff7-a9bf-896ee74b35ad', 'batch_id': 4, 'car_plate': 'WC 89', 'camera_id': 3, 'timestamp': '2024-01-01T08:01:18.835653', 'speed_reading': 94.5, 'producer_id': 'A'}
Batch 4 complete. Sleeping 0.335227s...

Batch 5 (1 records)

Published: {'event_id': '8d6827a6-5ca3-49ac-8d67-f01b64a31c4f', 'batch_id': 5, 'car_plate': 'KRN 7', 'camera_id': 3, 'timestamp': '2024-01-01T08:01:42.077427', 'speed_reading': 76.2, 'producer_id': 'A'}
...
Batch 383 (1 records)

Published: {'event_id': '516a5a31-d1b5-4ad7-bbe6-0ff0486d281b', 'batch_id': 383, 'car_plate': 'RT 3', 'camera_id': 3, 'timestamp': '2024-01-01T11:14:49.042629', 'speed_reading': 71.3, 'producer_id': 'A'}
Batch 383 complete. Sleeping 0.335227s...

Output for Streaming/Joining

```
Streaming query started.
Ensure producers A, B, and C are running before messages will arrive.
Interrupt the kernel (Kernel → Interrupt) to stop.

[batch 0] written=1 dropped=0 total_rows=1
[batch 1] written=1079 dropped=0 total_rows=1079
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=MU 6092 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 09:59:46 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=AYD 39 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 10:36:19 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=DN 7151 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 11:27:42 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=RM 4259 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 10:19:11 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=YXX 41 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 10:36:16 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
[batch 2] DROPPED – reason: unmatched A+B (cam 1 → cam 2) – no partner event within join window (45 seconds)
car_plate=IN 3520 camera_id_start=1 camera_id_end=None
timestamp_start=2024-01-01 10:50:16 timestamp_end=None
speed_reading=None avg_speed=None violation_type=AVERAGE
```

Chart Images

Chart 1

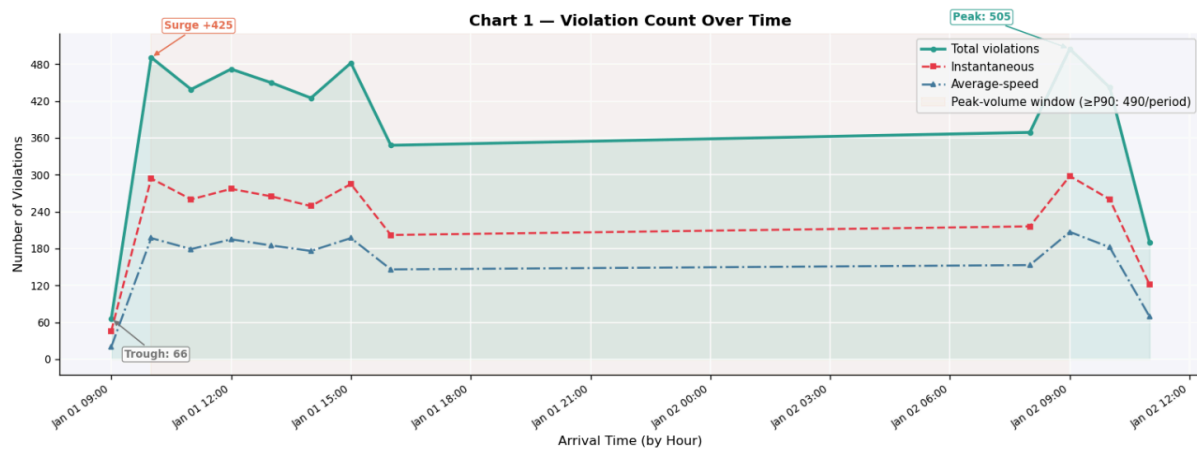


Chart 2

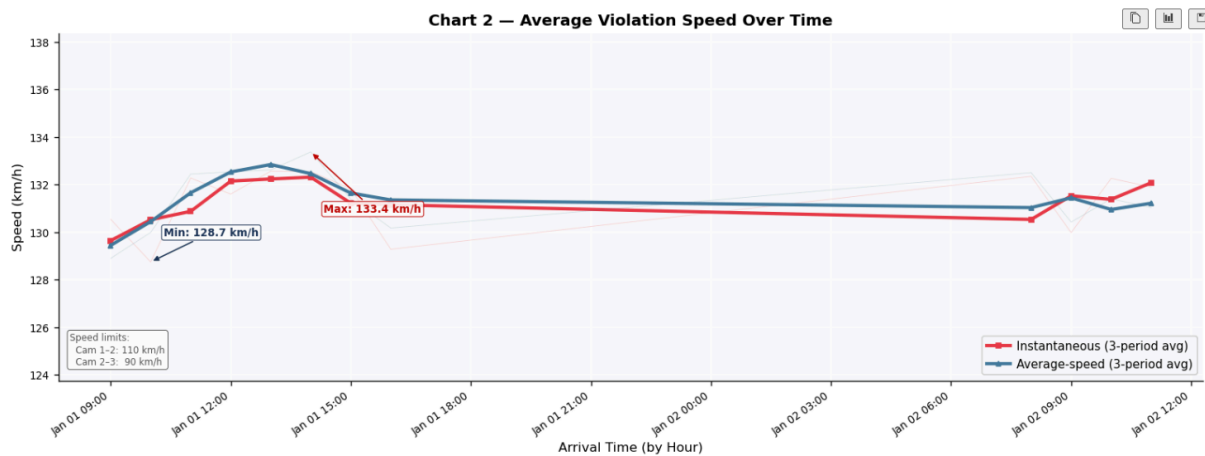


Chart 3

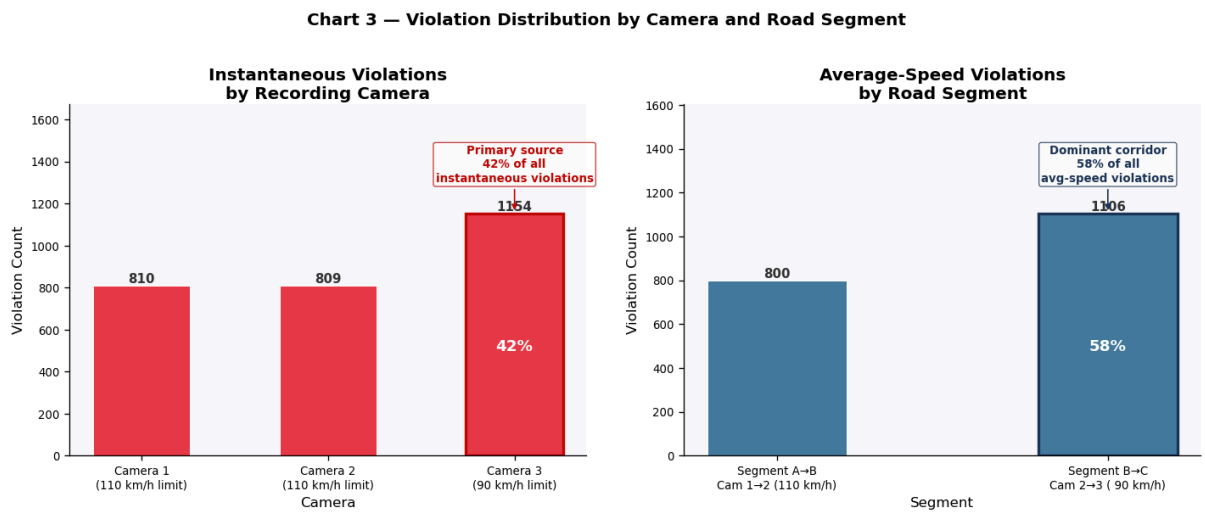


Chart 4

